

Design requirements for a commodity phased array radar

Status: **proposal and feasibility assessment**, 2026-10-02. Proposed yapnr demonstration; no hardware has been built or simulated for this study. All performance requirements are targets. Calculations are first-order estimates, and costs are allocations unless a price source is identified.

Decision: develop a **60 GHz FMCW radar with custom PCB antennas and feed networks**, using the external-antenna TI IWR6843 as the reference radio. A materials BOM below **\$100 at 10,000 units per year is plausible**. The proposed nominal allocation is **\$70**, with a **\$14 contingency**. The credible first application is low-speed autonomous ground vehicles and robots at approximately 0.5-20 m. A highway perception radar, reliable pedestrian detection in all conditions, and automotive qualification are separate projects.

This document follows yapnr's design-document structure and adds stable requirement IDs and verification criteria. The inspected main-branch snapshot is 8dfcf009e808208649387e00d6c3a053d9959ef2. No file literally named `rules_requirements` was present in that snapshot. The RF coupon and FEA design documents provide the closest local precedent. [R1-R4]

1. Goals and scope

1.1 What the demonstration must prove

The demonstration shall start with electrical and radiation-pattern requirements, select commodity components, synthesize copper RF structures, place and route the complete board, and evaluate the resulting coupled design. It shall show the physical manufactured radar and a reproducible comparison against a conventional design on the same stackup.

SYS-01 - Cost boundary. The recurring materials BOM shall be less than \$100 USD at a stated 10,000-unit annual purchase volume, with a design target of \$85 or less. It includes the radio, processing, memory, clocks, power conversion, interface components, PCB copper antennas, radome, connectors and mounting hardware. It excludes assembly, calibration labor, yield loss, certification, development equipment, engineering and host vehicle. These exclusions shall be priced separately; an external computer needed to perform radar processing must be included in the BOM. Verification: dated supplier quotations and an exact-part BOM at gate G5.

SYS-02 - Commodity construction. Use catalog ICs, ordinary SMT assembly and a commercially offered PCB process. No custom silicon, antenna-in-package radio, hand-tuned wire structures, exotic RF connector in each shipped unit, or external SDR is allowed in the demonstration BOM. A low-loss RF laminate is acceptable commodity hardware. Verification: manufacturing package and supplier process review.

SYS-03 - Application. The initial operational domain is a ground vehicle moving at up to 3 m/s, in a controlled test area, with radar targets at 0.5-20 m and radial target speeds up to 4.5 m/s. Initial environmental characterization covers dry conditions and 0-50 degrees C ambient. This is an additional perception sensor, not an approved sole braking sensor. Verification: a published test-domain declaration and recorded test runs.

SYS-04 - Scope of automation. yapnr shall generate and route every external RF interconnect, antenna feed and designated inverse-designed structure. The radio's internal PLL, mixers, LNAs and ADCs remain vendor-designed blocks with documented interface models. A successful result demonstrates the external signal chain and board design; it does not claim to have synthesized the radio silicon. Verification: generated geometry provenance from requirements to final board.

SYS-05 - Integration envelope. Target a sensor envelope no larger than 100 x 70 x 25 mm, powered from a regulated 5 V input with a 2 A source allowance, and no more than 4 W average consumption in the released 20 Hz profile. Output timestamp, mode, range, radial speed, azimuth, SNR/confidence and diagnostic status for each detection. Verify measured peak/average power, processing latency and junction temperature at the maximum declared ambient. An automotive battery/transient front end is outside this initial regulated-input domain and must be costed if added.

1.2 Non-goals for the first release

- Highway-speed autonomous driving, 100-200 m guaranteed detection, or dense 3D imaging.
- Automotive qualification or a system functional-safety claim inferred from the IC's qualification.
- Simultaneous fine angular resolution, narrow transmit beams and a large field of view from a handful of RF channels.
- A production-ready design generated solely from nominal FR-4 data.
- Replacing measured RF validation with KiCad DRC or replacing DRC with an electromagnetic result.

2. Frequency choice and feasibility

2.1 Candidate bands

The physical range resolution is approximately $\Delta R = c / (2 B)$, where B is the coherently observed bandwidth. The figures below are calculated ideal resolutions, before window broadening, interference and calibration error. They are not single-target accuracy guarantees. [R10]

Candidate	Representative usable bandwidth	Ideal range resolution	Assessment
902-928 MHz	26 MHz theoretical full-band span	5.77 m	Too coarse for the intended perception demo; large aperture
2.4 GHz	83.5 MHz US Part 15 band span	1.80 m	Better component access, still poor range separation
5.725-5.875 GHz	150 MHz span	1.00 m	Excellent intermediate RF-toolkit benchmark; coarse radar
24.0-24.25 GHz	250 MHz upper bound	0.60 m	Credible low-cost fallback; allow edge guards
60 GHz family	3.2 GHz observed sweep proposed	4.68 cm	Recommended combination of radar bandwidth and compact array
76-81 GHz	4 GHz example sweep	3.75 cm	Strong automotive path, but not an ISM-band answer

These spans describe engineering comparisons, not blanket permission to emit a particular waveform. The 2.4 and 5.8 GHz entries especially require waveform-specific regulatory review.

Terminology matters. The formal ISM designation near 60 GHz is **61-61.5 GHz**, not all of 57-71 GHz. The broader US 60 GHz region is available for qualifying unlicensed devices under specific rules. Restricting the radar to the 500 MHz ISM slice would give only 30 cm ideal resolution. Here, “60 GHz radar” means a design using the applicable unlicensed radar provisions, with the production band initially contained in **60-64 GHz**. [R8, R9]

At 915 MHz, half-wavelength spacing is 164 mm. An eight-element linear array has a 1.15 m end-to-end phase-center span; roughly a 3.3 m effective aperture is needed for a 5-degree broadside beam under

the usual uniform-aperture approximation. At 62 GHz, half-wavelength spacing is 2.42 mm and a 5-degree aperture is about 49 mm. Inverse design can improve efficiency, matching and packaging, but cannot remove this aperture requirement. [R11; calculations]

2.2 Radio selection

ARCH-01 - Reference radio. Use an external-antenna **IWR6843**, with the exact purchasable suffix and silicon revision frozen at G1. Its documented capabilities include 60-64 GHz operation, three TX channels, four RX channels, 6-bit TX phase control, complex receive conversion and integrated processing. TI's published comparison lists approximately **\$18.963 at 1,000 units**; this is a price anchor, not a 10,000-unit quote. The BOM allocates \$22. [R5-R7]

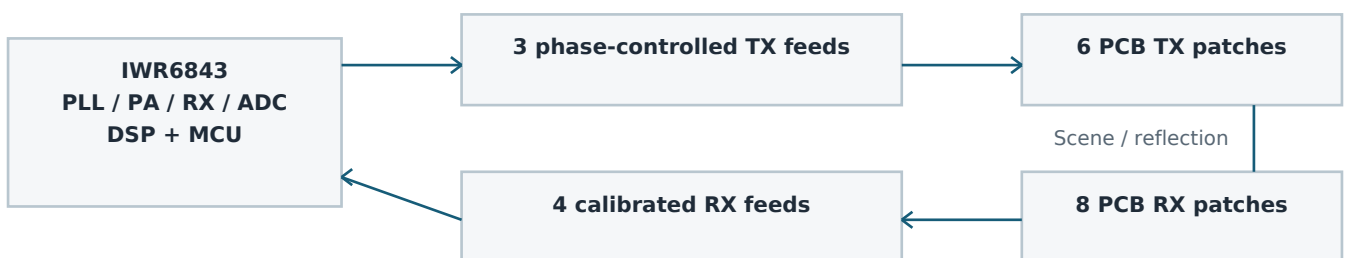
Use the IWR6843 because it supports an actual transmit phase-steering experiment as well as receive beamforming. The lower-cost IWR6432 is a viable digital/MIMO radar alternative, but its binary TX phase control is not a substitute for arbitrary phase steering. The newer IWR6844 also documents binary phase control; more channels alone do not establish equivalent beamforming capability. Do not silently substitute either part. [R15, R16]

24 GHz fallback. Infineon's BGT24LTR11 has accessible analog I/Q outputs, and the manufacturer demonstrates software-controlled FMCW. It is attractive for a 1D range demo or a switched-beam antenna. A true multibeam system additionally needs RF switching, a beam-forming network, baseband conversion and calibration; its one TX and one RX do not create a phased array by themselves. This fallback is useful if 60 GHz fabrication/metrology fails the early gates, with the range-separation requirement explicitly relaxed to approximately 1 m for a guarded sweep. [R17]

Sub-GHz conclusion. An SX1262 may supply an experimental CW source, but its normal packet receiver does not expose the coherent multi-channel sample stream required here. Its 500 kHz LoRa bandwidth corresponds to roughly 300 m ideal radar resolution if used as the observation bandwidth. A separate coherent receiver and stepped-frequency processing would be a different radio architecture. The revised baseline therefore has no LoRa dependency; telemetry can be added separately if needed. [R18; calculation]

3. Architecture and modes

3.1 External signal chain



Generated copper: launches, matching, corporate dividers, interconnects and radiators. Not to scale.

Power input -> protection and regulators -> reference clock and IWR6843 -> three phase-controlled TX ports -> synthesized launch/matching/feed structures -> three PCB antenna columns -> scene -> four PCB receive columns -> four synthesized feeds -> radio LNA/mixer/ADC -> onboard range/Doppler/angle processing -> timestamped detections over UART or CAN-FD.

Each column initially contains two vertically arranged radiating patches fed by a corporate divider. That is **six TX patches and eight RX patches**, with three and four independently controlled phase centers, respectively. The optimizer may change the radiator shape while preserving port contracts and the declared phase-center geometry. A patch count is not an independent-channel count.

ARCH-02 - Two explicit modes. Implement:

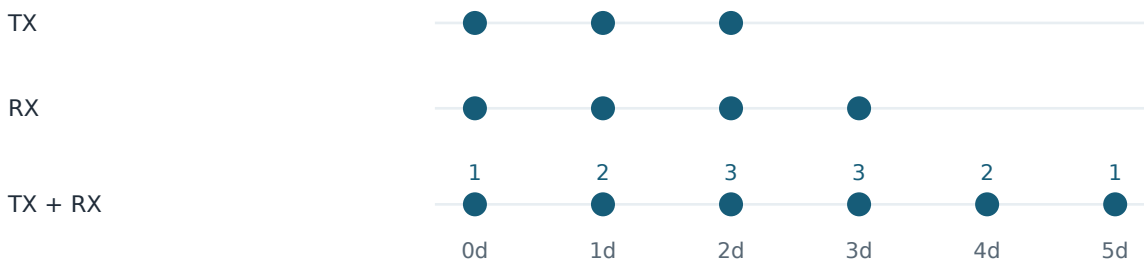
1. **Perception mode:** time-division multiplex the three TX channels, receive all four channels, estimate range and Doppler, compensate target motion between TX slots, then form angle estimates from the actual virtual-array geometry.
2. **Phased-transmit demonstration mode:** excite all three TX channels coherently, set phase progressions for -30, 0 and +30 degrees, and collect four RX channels. Measure the far-field beam shift and show the effect on target illumination. Each beam gets a separate coherent processing interval.

These modes shall have separate performance reports. They do not simultaneously deliver the same MIMO aperture, Doppler resolution and transmit gain. Verification: firmware configuration plus measured mode-specific patterns and detections.

ARCH-03 - Three-TX supply mode. The board shall implement the documented 1 V RF/PA LDO-bypass arrangement for three simultaneous TX, including the required PA supply connection, sequencing, current capacity and ripple limits. Budget for the documented 2.5 A peak on that rail; average board power is a different quantity. Verify the exact revision's data sheet and firmware controls before schematic release. Verification: schematic review and rail-current/ripple measurements in the worst transmit state. [R6, sections 7.6-7.7 and 8.3.1]

3.2 Phase-center geometry

Azimuth phase centers and virtual-position multiplicity



$d = 2.342$ mm. Twelve channel pairs occupy six distinct azimuth positions; translations omitted.

ANT-01 - Steering geometry. Start with azimuth TX positions 0, d , $2d$ and RX positions 0, d , $2d$, $3d$, where $d = c / (2 \times 64 \text{ GHz}) = 2.342$ mm. TX and RX banks may have a common physical translation and vertical separation. Their azimuth axes must remain parallel. The spacing is referenced to the highest design frequency. Verification: exact exported phase centers and full-wave embedded-element patterns.

This deliberately favors clean transmit steering. The twelve TX/RX pair measurements have only **six distinct azimuth sum positions**, with multiplicities 1, 2, 3, 3, 2, 1. They are not twelve independent uniformly spaced virtual elements. Equalizing the multiplicities affects noise; angle processing shall use the actual covariance and measured manifold. The expected resolution is coarse: roughly 18 degrees ideal broadside beamwidth for a uniformly weighted six-position equivalent, with a **25-degree two-target acceptance target**.

Spacing TX phase centers by $4d$ would instead produce twelve distinct MIMO positions with the same four RX positions. That is a separate layout option. Its sparse transmit array creates grating-lobe problems when all TX transmit coherently, so it cannot inherit the baseline's steering claims. The design shall never advertise both benefits without demonstrating how it achieves them. [R11; geometry calculation]

ANT-02 - Radiation targets. Across 60.3-63.8 GHz, require return loss of at least 10 dB per external antenna port, nominal per-column realized broadside gain of at least 5 dBi, and useful azimuth coverage of at least ± 45 degrees. In phased-transmit mode, main-lobe pointing shall be within 5 degrees of each commanded direction after calibration. No unintended lobe within the forward scan sector may exceed the selected main lobe. Treat total main-beam efficiency, cross-polarization, sidelobes and scan loss as measured outputs, not just antenna matching. Verification: embedded-element simulation and chamber patterns with the final radome.

ANT-03 - Honest angular output. Report single-target angle accuracy separately from two-target resolution. The baseline has no elevation-resolution requirement. Do not generate a convincing-looking 3D point cloud by interpreting an unmeasured elevation coordinate as data.

4. Waveform and performance requirements

4.1 Calculated starting profile

The following is a mutually consistent starting calculation, not a validated firmware configuration. Chip timing, analog-filter settings, quantization, memory layout and legal emissions must be checked at G1-G3. [R6, R10]

Quantity	Proposed value	Consequence
Chirp RF start	60.3 GHz	Lower edge guard
Ramp slope	50 MHz/us	Range-to-beat-frequency conversion
RF ramp duration	70 us	Ramp ends at 63.8 GHz
ADC start after ramp start	6 us	ADC observes 60.6-63.8 GHz
ADC capture	512 complex samples at 8 MS/s	64 us capture; 3.2 GHz observed bandwidth
Chirp repetition time	80 us	10 us reset/idle allowance
Perception sequence	20 cycles of TX0, TX1, TX2	60 chirps; 4.80 ms burst
Frame period	50 ms	20 frames/s; 45.20 ms between bursts
Per-TX sampling interval	240 us	About +/-5.0 m/s unambiguous radial velocity at 62 GHz
Per-TX coherent observation	About 4.80 ms	About 0.50 m/s nominal Doppler bin width
Beat frequency at 20 m	6.67 MHz	Requires a verified complex IF passband beyond this frequency
Raw radar cube	512 x 60 x 4 x 4 bytes	491,520 bytes before scratch/output storage

$\Delta_R = 4.68$ cm is the unwindowed range-bin scale. Windowing broadens peaks, so the acceptance requirement is 15 cm separation, not 4.68 cm. If the selected IF mode cannot pass 6.67 MHz with margin, reduce slope and recalculate duration, sample rate, samples, velocity ambiguity and duty together. Zero-padding does not recover missing bandwidth.

The raw cube is 480 KiB and is substantial relative to the chip's L3 storage. **DSP-01:** the firmware shall demonstrate a bounded memory plan including range/Doppler transforms, calibration arrays and output buffers. Streaming or in-place processing is expected. Meeting the RF arithmetic without fitting the processing workload is a failure.

For the phased-transmit demo, use three blocks of 20 chirps, one block per pointing direction, within the same burst budget. Each block lasts 1.60 ms: its nominal velocity-bin width is about 1.51 m/s. This mode does not inherit perception mode's velocity-resolution target.

4.2 Acceptance targets

All tests use the released firmware, final antenna/radome and a documented calibration. A “known-RCS target” means a characterized reflector with aspect and frequency dependence recorded, not an assumed human cross section.

ID	Requirement	Verification
RAD-01	Detect a 1 square meter reference target from 1-20 m at boresight	At least 90% frame detection over 200 frames per test point; 95% confidence interval reported
RAD-02	Detect a 0.1 square meter reference target from 1-10 m	Same criterion; pedestrian tests reported separately
RAD-03	Establish usable operation down to 0.5 m	Leakage/clipping and near-range filter test; publish any blind interval
RAD-04	Range error at most 10 cm for an isolated target at 1-20 m	95th-percentile absolute error, calibrated ground truth
RAD-05	Separate equal-strength targets 15 cm apart in range	At 5-10 m, matched angle/Doppler, two peaks with at least 3 dB valley; at least 90% of frames
RAD-06	Perception-mode velocity error at most 0.25 m/s; two-target velocity separation target 1.0 m/s	Controlled moving target; report window broadening and ambiguity limits
RAD-07	Angle error at most 3 degrees for an isolated target within +/-30 degrees	Post-processing SNR at least 15 dB; 95th-percentile absolute error
RAD-08	Resolve two equal-strength targets 25 degrees apart around boresight	Same range/Doppler; report the two-peak criterion and success rate
RAD-09	At least 20 perception frames/s and at most 100 ms output latency	End-to-end timestamps, including processing and transport
RAD-10	Fewer than one false confirmed track/minute in a specified empty scene	At least 30 min per scene; detector threshold and track policy frozen beforehand

RAD-01 and RAD-02 are boresight targets; a range-versus-angle map shall characterize scan degradation. Targets are not guaranteed throughout the full field of view. Separate trials shall cover walls, ground clutter, vehicles, foliage, pedestrians, wet radome and other radars. Passing the reference-target tests does not establish an all-weather vehicle-perception claim.

4.3 First-order sensitivity check

Use the monostatic approximation:

$$Pr = EIRP * Gr * \lambda^2 * \sigma / ((4\pi)^3 * R^4 * L).$$

For an illustrative conservative indoor profile, assume 12 dBm EIRP, 5 dBi realized gain at one RX port, 62 GHz, a 1 square meter target at 20 m, and 3 dB additional two-way environmental/radome loss. The received power is approximately **-117.3 dBm per receiver**. Feed loss is already included in realized gain and is not subtracted a second time.

With an assumed 12 dB receiver noise figure and Hann-window equivalent noise bandwidth $1.5/64 \text{ us} = 23.44 \text{ kHz}$, thermal noise per range bin is approximately **-118.3 dBm**. Ideal coherent accumulation of 20 chirps adds 13.0 dB, and combining four RX channels adds up to 6.0 dB. After a further assumed 3 dB processing/window allowance, the illustrative result is **17.0 dB SNR**. This deliberately credits only one TX's coherent sequence; extra duplicate MIMO measurements are not counted as free aperture gain.

At 10 m, a 0.1 square meter target gives approximately **19.0 dB** under the same assumptions. Halving range adds about 12 dB; reducing RCS by ten times costs 10 dB. These estimates support a prototype, not a detection guarantee. Leakage phase noise, interference, clutter, scan loss, nonideal coherent integration and target fluctuations can dominate. Gate G3 replaces the assumed loss and noise terms with measurements.

5. RF copper and fabrication requirements

RF-01 - Stackup as a contract. Freeze a supplier-specific stackup, RF laminate, thickness distribution, copper finish/roughness, etch tolerance, mask openings and via geometry. Start with a thin low-loss RF layer over a solid reference plane and ordinary lower layers for power/digital routing. Short-path characterized FR-4 is a cost-down experiment, not an interchangeable substitute. TI's RF layout guide provides a manufacturing reference; exact dimensions must come from this design's stackup. [R12]

RF-02 - Copper components. Synthesize at least two meaningful classes: the antenna radiator/column and a distributed feed component such as a two-way divider plus matching transition. Their emitted copper shall be the copper fabricated. Optimizing a picture, then manually redrawing it, is insufficient. Each structure shall carry ports, reference planes, frequency masks, a fabrication envelope and versioned S-parameters.

RF-03 - Feed budgets. From radio package launch to the declared antenna-feed plane, target no more than 1.5 dB total dissipative/mismatch loss per path, at least 10 dB return loss, and no more than 0.75 dB amplitude imbalance between equivalent paths. Target no more than 5 degrees residual channel phase error after calibration. A two-way divider's ideal -3.01 dB split is not dissipative loss; report excess loss separately. Verify across band, process corners and temperature samples.

RF-04 - Phase and delay. Specify electrical phase and group delay at common de-embedded reference planes, not only equal physical trace lengths. At 62 GHz and illustrative effective permittivity 2.8, the guided wavelength is about 2.89 mm; a 5-degree phase error corresponds to only **40 micrometers** of electrical path mismatch. Include bends, launches, vias, coupling and dielectric variation. Frequency-dependent calibration cannot repair lost signal power.

RF-05 - Microstrip and stripline. Prefer microstrip or grounded coplanar waveguide for short millimeter-wave feeds. If stripline is used, include both transitions and its ground structure in the full-wave model. Add a same-stackup stripline transition coupon as an optional toolkit benchmark; the radar shall not acquire a long lossy buried feed just to exhibit every primitive.

RF-06 - Manufacturability. Minimum width, gap, island size, via spacing and mask registration shall come from a quoted process. Use eroded, nominal and dilated geometries and sampled material/process variations during optimization. Do not retain sub-resolution features because a solver mesh happened to permit them. Initial acceptance requires a manufacturable design that meets hard masks in declared bounded corners; no fabricated statistical yield percentage shall be inferred from synthetic corners alone.

RF-07 - Coupling and leakage. Solve the antenna banks, feed region and representative nearby board/radome structures together. Include embedded-element patterns, active input match, port coupling and polarization. Require at least 10 dB input headroom below the applicable compression/blocker limits and at least 6 dB ADC headroom in the declared nearest-target/leakage test. Require no more than 3 dB measured desensitization from self-leakage at the acceptance ranges. Select antenna separation/shielding from that budget; an arbitrary isolation figure alone is not a radar validation.

RF-08 - Measurement access. Provide removable panel coupons and calibrated millimeter-wave probe or waveguide-fixture access to feed structures. A 6 GHz hobby VNA and SMA coupon cannot validate a 60 GHz product. Require a booked measurement path covering the product band before G2. Bench equipment is development cost and shall not be hidden in the \$100 claim.

6. yapnr workflow and acceptance contracts

6.1 Repository state and required extensions

At the inspected snapshot, coupon tooling exists on main, and its documentation explicitly says the boards have not been ordered or measured. Its supplied predictions cover 0.1-18 GHz, with stated specifications over 1-12 GHz. Those fits cannot be extrapolated into a validated 60 GHz material model. [R2]

Draft PR #30 reports that a divider passes its three-grid checks; antenna synthesis, a Wilkinson-type combiner and other cases remain unresolved. Issue #32 proposes catalog-part/distributed-element co-simulation and iteration with PnR. These are foundations, not evidence of an already qualified end-to-end millimeter-wave flow. [R3, R4]

PNR-01 - Declared RF objects. The proposed interface shall represent a copper RF component as geometry plus ports, reference planes, ground/keepout ownership, an orientation-dependent radiation model where relevant, fabrication limits and validation artifacts. These are proposed contracts; this document does not invent an existing yapnr API.

PNR-02 - Hierarchy. Preserve each TX/RX phase-center lattice as a constrained group. Place the radio, supplies, memory and interfaces with their own electrical/thermal contracts. Route RF connections using qualified line and transition families. Preserve ground returns and antenna clearances through fill, repair and final export.

PNR-03 - Coupled iteration. Evaluate compact network models during search; re-extract the changed RF region after placement/routing; run full-wave sign-off on the final exported copper. Coupled adjacent blocks require a joint model; blindly cascading independent two-ports misses coupling. Changes to stackup, ground fill, radome or nearby metal invalidate affected caches.

PNR-04 - Optimization priorities. Reject DRC, connection, fabrication, RF-mask, power and regulatory violations first. Among feasible candidates, minimize recurring cost, RF excess loss and area, then runtime. Selection is mechanical and seed-controlled, following the repository's Monte Carlo/successive-halving policy. An agent shall not hand-select a visually attractive board or manually route a failing design to claim automatic success. [R1]

PNR-05 - Evidence levels. Label outputs as analytical estimate, solver prediction, synthetic recovery, measured coupon or measured system. Require mesh refinement, passivity/energy checks, calibrated material loss and an independent solver or measured comparison. A radiated-fraction objective alone cannot establish antenna gain: loss and radiation must be distinguished. Report dispersion, finite-conductivity and roughness limitations of the solver at 60 GHz.

PNR-06 - Reproducibility. Hash the requirements, parts/models, stackup posterior, solver version, seed, geometry and firmware. Every solver worker has a timeout and resource budget. New engine behavior remains behind a default-off flag until its declared A/B result passes. Native KiCad DRC must report zero unconnected items and zero unwaived violations; RF acceptance remains an additional check. [R1]

6.2 End-to-end sequence

1. Freeze the operational domain, jurisdiction, part revision and RF ports.
2. Build the conventional reference and measurement fixtures; characterize same-process coupons.
3. Fit material/process parameters with uncertainty over the actual product band.
4. Optimize single radiator, two-way feed and launch primitives against robust masks.
5. Assemble TX/RX arrays and calculate their coupled embedded patterns.
6. Place and route the complete board; re-extract changed structures and iterate.
7. Sign off the exported board, radome, power, memory and waveform constraints together.

8. Build a small lot, calibrate, measure and compare against the reference with frozen analysis.

DEMO-01 - Fair comparison. Manufacture conventional and optimized variants with the same radio, stackup, outline constraint, firmware and measurement setup. Compare return loss, total efficiency, feed loss, pointing, angular response, detection probability, power, BOM and optimizer runtime. The optimized design passes if it meets the same hard requirements and demonstrates at least one preregistered gain, such as 20% smaller feed area or 0.5 dB less excess loss, without degrading the other hard limits. Both thresholds are proposed targets, not predicted wins.

DEMO-02 - Complete deliverables. Release the requirements, part-selection record, schematic/netlist, generated copper, native board, DRC, fabrication files, costing, model provenance, calibration data, firmware and raw test data. Include a live range-angle view and a measured beam-steering plot. The demonstration must show a real target, not only simulated S-parameters.

7. Spectrum and operating profiles

REG-01 - Initial US profile. Design for 47 CFR 15.255(c)(2)(iii)(A): its radar provisions specify a 14 dBm peak EIRP ceiling and at least 25.5 ms of qualifying transmitter off-time in every contiguous 33 ms interval; qualifying continuous off intervals are at least 2 ms. The outdoor fixed/qualifying vehicular alternative in (B) allows 20 dBm with a different off-time constraint. The baseline targets **12 dBm peak EIRP** for margin. A higher-power outdoor profile is optional and needs its own verification. [R8]

The proposed 4.80 ms burst every 50 ms leaves at least 28.20 ms outside the burst in any 33 ms window. At most one burst can intersect such a window. Even if a window boundary leaves a silent fragment shorter than 2 ms and that fragment is excluded, more than 26.20 ms remains, exceeding the 25.5 ms requirement. This conservative calculation treats the entire burst as transmitting and does not credit short inter-chirp gaps. Firmware shall enforce this envelope, including startup, calibration and diagnostic emissions; measured timelines and the certification test method must confirm it at G1.

REG-02 - Full-array EIRP. Verify the worst pointing direction, frequency, process and radome condition for every phase code. Three coherent equal TX ports can add $20 \log_{10}(3) = 9.54$ dB relative to one port at the same per-port power and element pattern. The power setting must back off accordingly. Beamforming improves spatial selectivity; it is not permission to exceed the radiated limit. Factory calibrations and startup states must obey the same envelope.

REG-03 - Product boundary. Review emissions, exposure, certification, installation conditions and the rule's restriction on external phase-locking inputs before hardware release. The baseline is one radio IC with an internal reference and an integral array; it shall not expose an external synchronization input intended to aggregate independently certified transmitters. Qualification of this architecture is a G1 regulatory deliverable. [R8, R13]

REG-04 - Regional variants. Do not assume US settings are legal in Europe, Japan or elsewhere. Each release shall name the country, applicable standard/rule version, waveform, power and installation category. The 24 GHz fallback requires its own regional review. The high-power 61-61.5 GHz option trades away bandwidth and is not the default design.

8. Cost model

8.1 Recurring materials allocation

The following is a planning budget for the complete sensor, not a supplier-quoted BOM. PCB, radome and assembly figures are especially sensitive to process and volume. No volume discount below the cited radio price is needed to make the nominal arithmetic work.

Item	Quantity or boundary	Nominal USD	Planning range USD
IWR6843 external-antenna radio/processor	1; price anchor about \$18.96/1ku	22	19-28
Power conversion and filtering	All rails, 1 V bypass capability, inductors	8	6-12
Flash, 40 MHz reference, reset	One set	3	2-5
External passives and RF support	Decoupling, bias, damping; copper counted below	4	3-6
PCB including RF laminate and antennas	One controlled-process board	16	10-25
Interface and input protection	UART/CAN-FD, ESD, input protection	5	3-7
Connectors and mounting hardware	Shipped product hardware	4	2-6
Radome and enclosure	Commodity molded material	6	4-10
Thermal/spreader/shield allowance	As required by measured design	2	1-4
Materials subtotal		70	50-103
Design contingency	20% of nominal, not another physical part	14	
Budget including contingency		84	

The high end exceeds \$100: the constraint is plausible, not closed. A \$25 RF PCB and \$28 radio consume most of the margin. **COST-01:** if the quoted materials total is \$85 or more, review cost-down options before full-array fabrication. If the final qualified materials BOM is \$100 or more, the original demo requirement fails.

At nominal materials cost, illustrative assembly \$6, calibration/test \$5 and packout \$3 give \$84 before yield. At 95% yield, a simplified fully-at-risk model gives $\$84/0.95 = \88.42 per good unit. These are process allowances, not quotations; test fixture amortization, freight, tariffs and warranty remain outside that estimate. Never present the \$84 materials-with-contingency figure as a fully quoted shipped cost.

8.2 Development resources

Required resources include an RF PCB engineer, radar firmware/DSP work, yapnr solver/PnR work, 60-64 GHz network and antenna measurements, a motion/reflector test setup and at least two PCB iterations. Low unit cost does not imply low development cost. Shared or rented millimeter-wave facilities are a credible route, but access and pricing are unresolved.

A planning sequence is 2-3 weeks for architecture/profile proof, 3-6 weeks for coupons and supplier characterization, 4-8 weeks for primitives/array/PnR closure, and 4-8 weeks for two hardware iterations and system validation, with overlap. **12-24 weeks is a planning envelope for an experienced team with lab access**, not a commitment; immature solver work or fabrication lead times can extend it.

9. Verification gates and stop conditions

G0 - Requirements and radio access. Freeze the domain, reference part, source availability and exact pricing basis. Verify firmware can acquire the required complex data and control TX phase. Stop if the chosen part does not expose the required mode or the project can only access a closed antenna module.

G1 - Waveform and implementation proof. Demonstrate chirp/IF feasibility, memory fit, processing time, three-TX supply behavior, regulatory profile and complete RF-active timeline on a reference platform. Use the 20-cycle profile and verify the complete rolling-window off-time accounting. Stop if the system needs an unbudgeted external processor or cannot meet its legal power/timing envelope.

G2 - Material and primitive proof. Measure same-process lines, launches, a divider and a radiator over the operating band. Require model-to-measurement agreement within 0.5 dB magnitude and 5 degrees transmission phase for the specified feed structures after de-embedding, with resonance centers within 1%. If not, update the model or process before optimizing the complete radar. Access to appropriate metrology is a prerequisite.

G3 - Array and receiver proof. Validate coupled antenna patterns, calibrated steering, leakage headroom, IF response and the reference-target link budget. Run the perception and phased-transmit modes independently. Recalculate range claims from measured sensitivity and legal EIRP. Stop if the accepted geometry cannot support the promised angular performance.

G4 - Automatic layout proof. Generate and route complete reference/optimized boards using frozen contracts and mechanical selection. Require native DRC plus final-copper RF sign-off. Compare runs across a fixed seed set and fixed compute budgets; report failures as well as the winning candidate. No hand-edited copper rescue qualifies.

G5 - Built-system and cost proof. Characterize at least ten units across two fabrication lots, including conventional controls. Run RAD-01 through RAD-10, pattern tests and environmental characterization. Publish unit/lot variation; ten units do not prove mass-production yield. Close supplier quotes and report materials, assembly, calibration and development costs separately.

G6 - Demonstration release. Show requirements -> optimization -> full-board PnR -> predicted fields -> fabricated radar -> calibrated target measurements. Freeze source/configuration hashes and make datasets reproducible. A full vehicle qualification program begins only after this gate; it is not an implied deliverable of the demo.

10. Feasibility assessment

Hardware cost: plausible, medium-to-high confidence at concept level. A roughly \$19 catalog radar IC makes a sub-\$100 sensor credible without custom silicon. The uncertain cost drivers are RF PCB yield, packaging and calibration, rather than the transceiver alone.

Useful short-range radar: feasible in principle, medium confidence for these exact targets.

Available bandwidth and coherent reception support the proposed range/Doppler performance. Link-budget estimates have enough margin for controlled reference targets, but measured leakage and clutter decide useful performance. Human RCS varies strongly and is not guaranteed by the reflector result.

Phased transmit steering: feasible but modest. Three close-spaced TX phase centers can visibly steer a broad beam. They do not produce a pencil beam. The baseline intentionally accepts coarse angular separation; larger arrays or more channels would need another geometry and cost study.

End-to-end inverse design and automatic PnR: the largest project risk. The inspected implementation is still developing the required primitives, and current coupons are below the proposed band. At 60 GHz, etch, roughness, ground transitions, material dispersion and coupled radiation are first-order effects. Meeting this document requires new validation work, not simply retuning a 5.8 GHz example.

Autonomous-vehicle suitability: credible for a constrained robotics demonstration; unproven for road deployment. Long range, fine angle resolution, interference resilience, adverse weather, safety diagnostics and qualification are not established by a cheap BOM or a good laboratory plot.

Recommended next commitment: fund G0-G2 before the full radar. In parallel with the 60 GHz system definition, retain the repository's existing 5.8 GHz work as a faster way to stabilize inverse-designed feeds, co-simulation and PnR integration. Proceed to the complete 60 GHz board only after measured primitive closure and an RF fabrication quote within budget.

11. Sources and provenance

Manufacturer and regulatory pages were reviewed on 2026-10-02. Prices are indicative public figures and may change. Equations, architecture choices, targets, allocations and timing examples in this document are the author's engineering analysis; they are not vendor performance promises. No full-wave solve or physical measurement was performed for this assessment.

- **R1.** yapnr agent and engineering rules at the inspected commit.
- **R2.** RF coupon design and coupon user guide. Establish current frequency coverage and unmeasured status.
- **R3.** RF inverse-design draft PR 30 and issue 29. Live work status inspected separately from main.
- **R4.** RF topology toolkit issue 32 and FEA design.
- **R5.** TI IWR6843 product documentation.
- **R6.** IWR6843 and IWR6443 data sheet Rev F. RF mode, supply, memory and timing constraints.
- **R7.** TI IWR6843 and IWR6843AOP comparison and 1ku pricing. Published price anchor; exact variant quote remains open.
- **R8.** 47 CFR 15.255. US unlicensed 60 GHz radar provisions, current text displayed through 2026-09-30.
- **R9.** ITU spectrum-management training on ISM allocations. Includes 24-24.25 and 61-61.5 GHz allocations; national device rules remain separate.
- **R10.** TI Programming Chirp Parameters in Radar Devices. Range, velocity and sampling relationships.
- **R11.** Analog Devices phased-array characteristics and grating lobes and beam squint.
- **R12.** TI RF PCB Design Manufacturing and Validation Guide.
- **R13.** TI Radar Device Regulatory Compliance Overview Rev D. Secondary engineering guidance; the governing rule controls.
- **R14.** 47 CFR 15.249 and 47 CFR 15.247. Alternative-band regulatory context.
- **R15.** TI IWRL6432 data sheet and comparison pricing.
- **R16.** TI IWRL6844. Newer alternative; binary phase-control distinction.
- **R17.** Infineon Distance2GoL application note AN615. Evidence of a commodity 24 GHz analog-I/Q FMCW path.
- **R18.** Semtech SX1262 documentation and manufacturer SX126x driver interface. Packet-radio and bandwidth context.